

6.7 PreQuiz: Polynomials (no calculator)

1. [Maximum mark: 7]

Let $f(x) = x^2 + 2x - 15$.

(a) Write $f(x)$ in factored form, such that $f(x) = (x - a)(x - b)$. [2]

(b) Write down the x -intercepts of the graph of f . [1]

(c) Write down the y -intercept. [1]

(d) Write $f(x)$ in vertex form, i.e. $f(x) = (x - h)^2 + k$. [1]

(e) Find the coordinates of the vertex of the parabola. [2]

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2. [Maximum mark: 7]

Consider the function $g(x) = -(x - 4)^2 + 9$.

- Write down the coordinates of the vertex. [1]
- Write down the equation of the axis of symmetry. [1]
- Show that $g(x) = -x^2 + 8x - 7$. [2]
- Solve $g(x) = 0$ to find the x -intercepts. [3]

3. [Maximum mark: 7]

The line $y = 2x$ and the parabola $y = x^2 - 3$ are drawn on the same axes.

(a) Show that the x -coordinates of the points of intersection satisfy

$$x^2 - 2x - 3 = 0. \quad [2]$$

(b) Show that $x^2 - 2x - 3 = 0$ has two distinct real roots. [1]

(c) Factorise $x^2 - 2x - 3$. [1]

(d) Find the x -coordinates of the points of intersection. [1]

(e) Find the coordinates of both points of intersection as (x, y) pairs. [2]

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4. [Maximum mark: 7]

The equation $2x^2 - kx + 8 = 0$ has two equal roots.

(a) Write down the discriminant in terms of k . [1]

(b) Find the values of k . [2]

(c) For the positive value of k , find the repeated root. [2]

(d) State the values of k for which the equation has no real roots. [2]

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